

WiFi Ambient Synth

Listen to the invisible.

WiFi Ambient Synth is an explorable ambient synthesizer driven by the wireless environment. Networks become presences on a terminal star map; moving toward an object brings its voice into focus. A Rust engine generates network voices, drones, an acid line, gestures and packet-driven textures. The nine-channel Mix Lab lets the listener adjust the balance and compare two mix states.

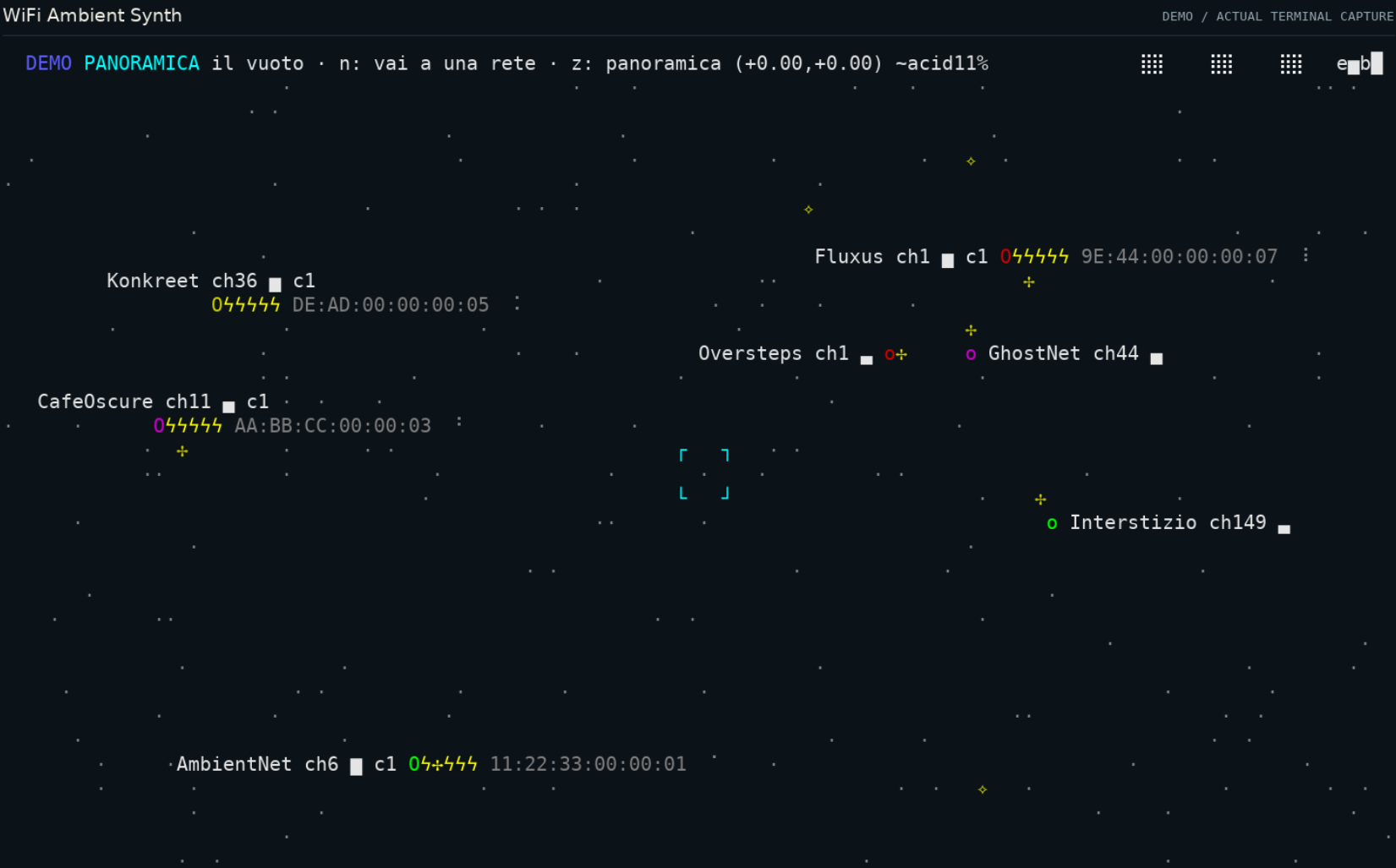
The project is made by Ettore, with Claude Code and Codex assisting development and review. It is an experimental instrument, version 0.1.0, distributed under GNU GPL version 3.

A small story

The first surviving source dates to August 2026. As the instrument grew into a star map, the work shifted from turning events into notes toward exploring what attention can do to a sound field. Layer recordings, real traffic replay and kernel timestamps exposed mistakes that a compelling picture could have hidden. The result is a project whose visual presentation follows what the instrument actually does.

EXPERIMENTAL 0.1.0 / RUST / GPLv3

A field you can explore



Overview of simulated networks. Actual running terminal app, DemoSource; no live RF capture.

Give each layer its space

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DEMO / ACTUAL TERMINAL CAPTURE

MIX LAB

```
> voci ambientali      +0.0 dB  attivo
acid/bassline          -9.0 dB  attivo
drone                  -3.0 dB  attivo
witness voice          -5.0 dB  attivo
gesti/eventi           -6.0 dB  attivo
click ritmico          -6.0 dB  attivo
databit (pacchetti)    -6.0 dB  attivo
riverbero              -2.0 dB  attivo
master                 +0.0 dB  attivo
```

A: vuoto · B: vuoto

? aiuto Tab mappa ↑↓ canale ↔ dB m mute s solo r reset a/b salva A/B richiama

Nine channels with gain, mute, solo and A/B mix snapshots. Actual running app, DemoSource.

Start listening

Try the hardware-free demo with Rust installed:

```
cargo run --locked --release -- --demo
```

Name: WiFi Ambient Synth. Binary and repository: wifi-ambient.

Creator: Ettore / ettword. Development assistance: Claude Code and Codex.

Status: experimental, 0.1.0; no official packaged desktop release yet.

Platforms verified: Windows and Linux demo; Linux/WSL2 monitor capture on Atheros AR9271.

Live capture: Linux wifi build, compatible monitor adapter and capture permissions.

Interface: Unicode terminal star map, contextual help and nine-channel Mix Lab.

License: GNU GPL version 3 only, without warranty; dependencies retain their licenses.

Home and source: <https://github.com/ettword/wifi-ambient>

Project contact: <https://github.com/ettword>

Media and use

The media directory contains the project mark, wordmark, cover, social card, four demo captures and an original 30-second demo audio excerpt. The captures contain synthetic network names. The audio is generated by the instrument; no reference-album material is included. Source and generated project assets are distributed under the repository's GPLv3 license. The project is not affiliated with the artists discussed in its research notes.

Adaptive radio attention is the default following AR9271/WSL tests, with a legacy fallback. Burst-driven sound experiments remain opt-in. Real-time replay is not a sample-exact offline renderer. The instrument is still experimental.

github.com/ettword/wifi-ambient